

ST CHARLES COUNTY AP, MO, USA

WMO: 724347

Lat: 38.929N

Lon: 90.428W

Elev: 436

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	11.4	-3.9	4.5	9.0	0.3	5.7	13.2	26.5	33.0	24.5	32.5	8.0	320	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	95.6	76.8	92.8	76.1	90.4	75.2	79.9	89.6	78.4	88.2	77.2	86.8	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	144.5	84.7	75.5	135.6	83.3	74.3	130.3	82.2	43.6	89.4	42.2	88.3	40.8	86.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.8	20.2	18.0	-0.3	99.6	5.2	2.8	-4.1	101.6	-7.1	103.2	-10.0	104.8	-13.8	106.8	
			DB												
			WB	-1.2	82.1	5.0	1.9	-4.8	83.5	-7.7	84.6	-10.5	85.7	-14.1	87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	31.0	33.7	45.5	56.8	66.4	75.6	78.2	76.7	69.2	57.4	45.6	35.3
	DBStd	18.86	11.11	11.03	10.75	9.13	8.08	5.64	5.54	5.42	7.16	8.97	9.39	9.61
	HDD50	2003	595	467	217	38	1	0	0	0	29	191	465	
	HDD65	4711	1055	877	609	270	85	3	0	1	38	270	582	921
	CDD50	4220	5	10	76	243	510	769	874	829	576	259	60	9
	CDD65	1453	0	0	4	25	128	322	409	365	164	35	1	0
	CDH74	14530	0	1	52	336	1174	3200	4335	3493	1560	362	17	0
	CDH80	5980	0	0	4	83	363	1336	1969	1532	596	95	2	0
	WSAvg	7.8	9.0	9.1	9.6	10.0	8.1	6.5	5.5	5.1	5.9	7.6	8.4	8.6
	PrecAvg	41.6	2.4	2.2	3.1	4.6	5.0	4.4	3.8	3.4	3.3	3.2	3.5	2.6
Precipitation	PrecMax	58.1	8.4	5.0	6.9	11.0	11.1	11.5	8.0	6.1	10.2	13.1	7.4	11.1
	PrecMin	29.1	0.7	0.4	1.1	1.9	1.0	0.6	0.7	0.9	0.4	0.6	0.7	0.4
	PrecStd	7.9	1.7	1.2	1.3	2.2	2.6	2.6	1.9	1.4	2.1	2.4	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	70.1	79.6	87.3	91.1	96.4	101.2	99.0	94.0	87.8	76.0	66.4
		MCWB	57.1	57.4	62.7	67.8	72.7	75.3	76.0	76.9	73.9	71.9	62.9	59.9
	2%	DB	58.7	63.0	73.9	82.0	87.4	93.2	96.2	95.0	90.4	82.2	70.5	59.5
		MCWB	52.0	54.2	60.8	64.8	71.4	75.5	77.7	76.5	73.7	68.1	59.5	53.8
	5%	DB	52.7	56.5	68.2	77.8	84.0	90.8	92.7	91.4	87.1	78.2	65.6	54.5
		MCWB	46.7	49.7	57.8	62.5	69.6	75.2	77.4	75.6	71.8	65.2	58.1	49.6
	10%	DB	46.8	50.4	63.2	73.2	81.1	87.9	88.3	82.8	73.0	61.3	49.5	
		MCWB	42.3	45.1	55.6	61.0	68.3	73.6	76.1	74.6	69.4	63.3	54.2	45.3
	0.4%	WB	59.8	60.8	66.1	71.0	76.1	80.0	82.2	81.4	77.4	73.4	66.0	61.5
		MCDB	65.3	67.5	76.3	82.2	86.4	89.7	91.5	91.3	88.3	83.4	72.5	65.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.7	55.9	62.5	67.2	73.4	78.2	80.3	79.3	75.8	71.1	62.4	55.8
		MCDB	57.7	61.4	70.6	77.2	83.0	88.9	89.8	89.3	85.7	79.3	67.6	59.2
	5%	WB	47.6	50.4	59.9	64.9	71.7	76.7	79.0	77.7	74.3	68.3	58.9	50.4
		MCDB	51.9	55.9	66.8	74.4	80.9	87.1	88.4	87.4	82.4	74.4	64.5	53.5
	10%	WB	42.0	44.6	56.5	62.5	70.1	75.3	77.6	76.3	72.5	65.0	55.0	46.0
		MCDB	47.1	50.1	62.1	71.3	78.5	84.6	86.9	85.2	79.5	71.3	60.7	48.7
Mean Daily Temperature Range	5% DB	MDBR	17.1	17.8	20.2	21.4	19.8	19.7	19.1	20.2	22.2	21.4	19.6	15.7
		MCDBR	27.4	28.6	28.3	29.1	23.5	22.9	22.4	24.1	25.3	27.1	27.9	23.7
		MCWBR	20.1	20.5	17.0	14.9	10.7	9.0	8.0	9.0	9.9	13.7	18.2	18.2
	5% WB	MCDBR	25.4	26.2	24.6	24.3	20.4	20.2	19.7	21.1	21.0	21.8	23.3	21.3
		MCWBR	20.1	20.2	16.3	14.6	10.6	9.1	8.8	9.2	9.3	13.0	17.3	17.9
Clear-Sky Solar Irradiance	taub	0.304	0.315	0.346	0.401	0.440	0.446	0.476	0.464	0.411	0.353	0.310	0.300	
	taud	2.467	2.428	2.410	2.253	2.205	2.226	2.141	2.175	2.293	2.463	2.526	2.494	
	Ebn at Noon	277	289	289	278	268	266	256	256	264	272	273	270	
	Edh at Noon	25	30	34	43	45	44	48	45	38	29	24	23	
All-Sky Solar Radiation	RadAvg	642	874	1184	1562	1759	2006	1974	1757	1507	1073	726	541	
	RadStd	72	87	108	93	149	159	134	102	95	119	73	60	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page